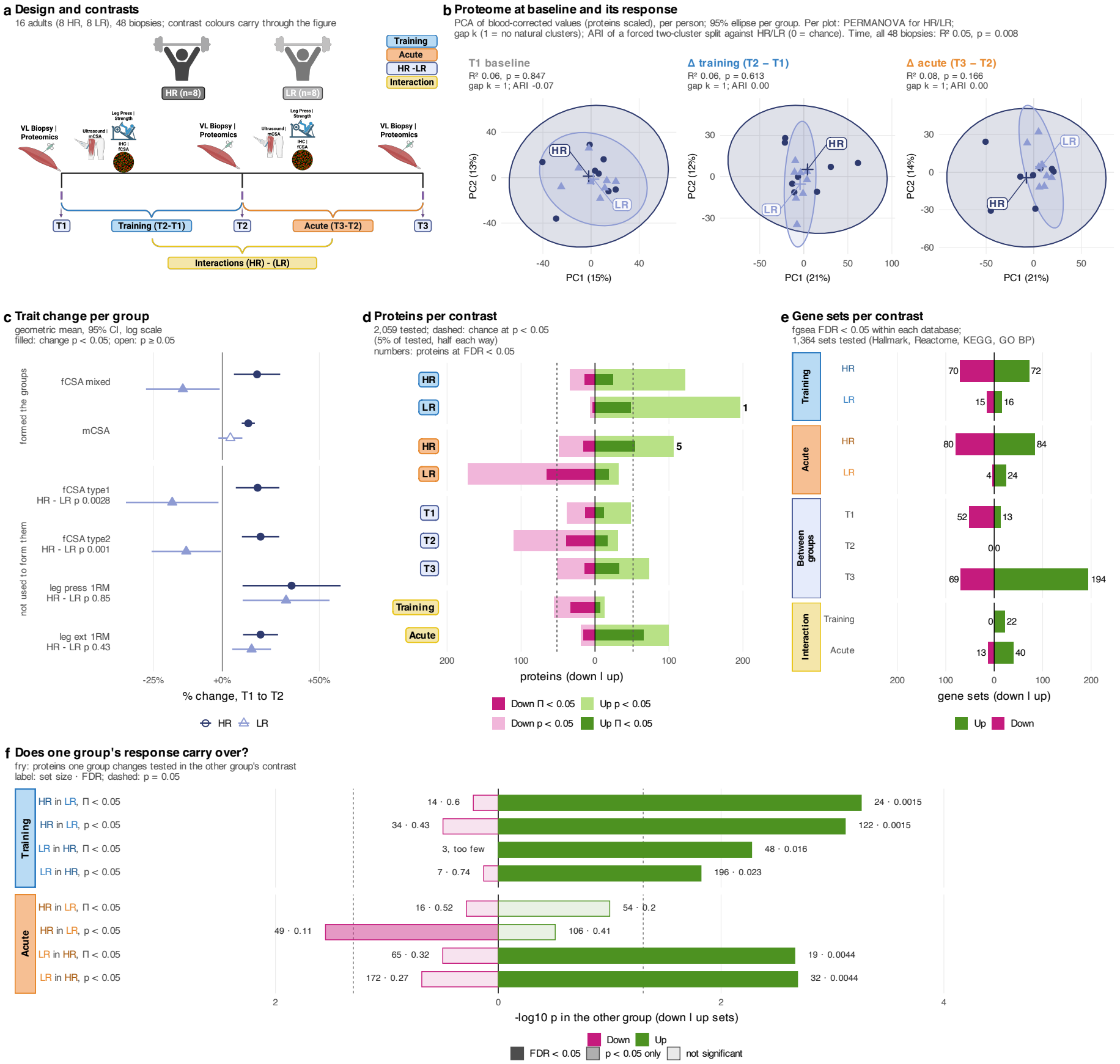

a Design and contrasts
16 adults (8 HR, 8 LR), 48 biopsies; contrast colours carry through the figure
Training
Acute
HR -LR
Interaction
HR (n=8)
LR (n=8)
VL Biopsy | Proteomics
Ultrasound | mCSA
Leg Press | Strength
IHC | fCSA
T1
Training (T2-T1)
T2
Acute (T3-T2)
T3
Interactions (HR) - (LR)
b Proteome at baseline and its response
PCA of blood-corrected values (proteins scaled), per person; 95% ellipse per group. Per plot: PERMANOVA for HR/LR;
gap k (1 = no natural clusters); ARI of a forced two-cluster split against HR/LR (0 = chance). Time, all 48 biopsies: R² 0.05, p = 0.008
T1 baseline
R² 0.06, p = 0.847
gap k = 1; ARI -0.07
Δ training (T2 – T1)
R² 0.06, p = 0.613
gap k = 1; ARI 0.00
Δ acute (T3 – T2)
R² 0.08, p = 0.166
gap k = 1; ARI 0.00
PC1 (15%)
PC2 (13%)
PC1 (21%)
PC2 (12%)
PC1 (21%)
PC2 (14%)
HR
LR
c Trait change per group
geometric mean, 95% CI, log scale
filled: change p < 0.05; open: p ≥ 0.05
formed the groups
not used to form them
fCSA mixed
mCSA
fCSA type1
HR - LR p 0.0028
fCSA type2
HR - LR p 0.001
leg press 1RM
HR - LR p 0.85
leg ext 1RM
HR - LR p 0.43
-25%
+0%
+50%
% change, T1 to T2
HR
LR
d Proteins per contrast
2,059 tested; dashed: chance at p < 0.05
(5% of tested, half each way)
numbers: proteins at FDR < 0.05
HR
LR
T1
T2
T3
Training
Acute
1
5
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
e Gene sets per contrast
fgsea FDR < 0.05 within each database;
1,364 sets tested (Hallmark, Reactome, KEGG, GO BP)
Training
Acute
Between groups
Interaction
70
72
15
16
80
84
4
24
52
13
0
0
69
194
0
22
13
40
gene sets (down | up)
Up
Down
f Does one group's response carry over?
fry: proteins one group changes tested in the other group's contrast
label: set size · FDR; dashed: p = 0.05
Training
Acute
HR in LR, Π < 0.05
HR in LR, p < 0.05
LR in HR, Π < 0.05
LR in HR, p < 0.05
14 · 0.6
24 · 0.0015
34 · 0.43
122 · 0.0015
3, too few
48 · 0.016
7 · 0.74
196 · 0.023
16 · 0.52
54 · 0.2
49 · 0.11
106 · 0.41
65 · 0.32
19 · 0.0044
172 · 0.27
32 · 0.0044
-log10 p in the other group (down | up sets)
Down
Up
FDR < 0.05
p < 0.05 only
not significant